Epic-FHIR — API category: Infrastructure Exchange (sections 24 – 24.3.3)

Source: Epic on FHIR (fhir.epic.com) documentation, mirrored 2026-11. Text-only edition for AI ingestion (NotebookLM); pages follow the source order (its table of contents when one exists).

24 API category: Infrastructure Exchange

File: site/apis/24-Infrastructure-Exchange/index.html

R4 (1)

API | Method | URL template | Type
Endpoint.Read (R4) | GET | /api/FHIR/R4/Endpoint/{ID} | Industry-Standard

STU3 (1)

API | Method | URL template | Type
Endpoint.Read (STU3) | GET | /api/FHIR/STU3/Endpoint/{ID} | Industry-Standard

Non-FHIR (3)

API | Method | URL template | Type
GetEncoderMessage | GET | api/epic/2016/Billing/Encoder/GETENCODERMESSAGE/EncoderMessage | Epic Public
HL7v2 | POST | api/epic/2015/EDI/HTTP/HL7V2/{AIP} | Interface
SetEncoderMessage | POST | api/epic/2016/Billing/Encoder/SETENCODERMESSAGE/EncoderMessage | Epic Public

24.1 Infrastructure Exchange: R4

File: site/apis/24-Infrastructure-Exchange/1-R4/index.html

* Endpoint.Read (R4)

24.1.1 Endpoint.Read (R4)

File: site/apis/24-Infrastructure-Exchange/1-R4/910-Endpoint-Read-R4.html
Epic API ID | 910
FHIR version | R4
Categories | Infrastructure Exchange
API type | Industry-Standard
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2019
HTTP method | GET
URL template | /api/FHIR/R4/Endpoint/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=910

Description

The FHIR Endpoint resource describes the technical details of a location that can be connected for the delivery or retrieval of information. The resource contains information on how to connect and for what purposes. This endpoint does not need to be the current system, it can describe locally hosted services, regional services, or national services.

Talk to the target organization for their supported protocols when using this resource.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Endpoint | Endpoint | False
Endpoint.address | String | False | | | The provider address. This is a direct address, not a general address.
Endpoint.connectionType | Coding | False | | | This is always "direct-project".
Endpoint.name | String | True | | | A name by which you can identify this endpoint.
Endpoint.payloadMimeType | String | True | | | This element returns "text/xml", "text/plain", and "application/pdf".
Endpoint.payloadType | CodeableConcept | False | Yes | | This element returns "urn:hl7-org:sdwg:ccda-structuredBody:1.1" and "urn:hl7-org:sdwg:ccda-nonXMLBody:1.1".

Endpoint.status | String | True | | | This is always "active".

Sample request

```
https://hostname/instance/api/FHIR/R4/Endpoint/entgyDkba56.PhVi9z2CS4.mlytOiVdVkC60AeMRB8JM3
```

Sample response


```
{
  "resourceType": "Endpoint",
  "id": "entgyDkba56.PhVi9z2CS4.mlytOiVdVkC60AeMRB8JM3",
  "status": "active",
  "connectionType": {
    "system": "http://terminology.hl7.org/CodeSystem/endpoint-connection-type",
    "code": "direct-project",
    "display": "Direct Project"
  },
  "name": "Epic User, MD",
  "payloadType": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/endpoint-payload-type",
          "code": "urn:hl7-org:sdwg:ccda-structuredBody:1.1"
        }
      ],
      "text": "Documents following C-CDA constraints using a structured body."
    },
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/endpoint-payload-type",
          "code": "urn:hl7-org:sdwg:ccda-nonXMLBody:1.1"
        }
      ],
      "text": "Documents following C-CDA constraints using a non structured body."
    }
  ],
  "payloadMimeType": [
    "text/xml",
    "text/plain",
    "application/pdf"
  ],
  "address": "mailto:epic@epic.com"
}
```


24.2 Infrastructure Exchange: STU3

File: site/apis/24-Infrastructure-Exchange/2-STU3/index.html

* Endpoint.Read (STU3)

24.2.1 Endpoint.Read (STU3)

File: site/apis/24-Infrastructure-Exchange/2-STU3/866-Endpoint-Read-STU3.html
Epic API ID | 866
FHIR version | STU3
Categories | Infrastructure Exchange
API type | Industry-Standard
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2018
HTTP method | GET
URL template | /api/FHIR/STU3/Endpoint/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=866

Description

The FHIR Endpoint resource describes the technical details of a location that can be connected for the delivery or retrieval of information. The resource contains information on how to connect and for what purposes. This endpoint does not need to be the current system, it can describe locally hosted services, regional services, or national services.

Talk to the target organization for their supported protocols when using this resource.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Endpoint | Endpoint | False
Endpoint.address | String | False | | | The provider address. This is a direct address, not a general address.
Endpoint.connectionType | Coding | False | | | This is always "direct-project".
Endpoint.name | String | True | | | A name by which you can identify this endpoint.
Endpoint.payloadMimeType | String | True | | | This element returns "text/xml", "text/plain", and "application/pdf".
Endpoint.payloadType | CodeableConcept | False | Yes | | This element returns "urn:hl7-org:sdwg:ccda-structuredBody:1.1" and "urn:hl7-org:sdwg:ccda-nonXMLBody:1.1".
Endpoint.status | String | True | | | This is always "active".

Sample request

```
https://hostname/instance/api/FHIR/STU3/Endpoint/efPfWtfd4gDD7HOVJBrm1GA3
```

Sample response

```
<?xml version="1.0" encoding="UTF-8"?>
<Endpoint xmlns="http://hl7.org/fhir">
  <id value="ehNSAQVtoxEih.WlA-M0vLnO86lyeQd5rX8BdjRileRk3"/>
  <status value="active"/>
  <connectionType>
    <system value="http://hl7.org/fhir/ValueSet/endpoint-connection-type"/>
    <code value="direct-project"/>
    <display value="Direct Project"/>
  </connectionType>
  <name value="Holiday, Doc"/>
  <payloadType>
    <coding>
      <system value="http://hl7.org/fhir/ValueSet/endpoint-payload-type"/>
      <code value="urn:hl7-org:sdwg:ccda-structuredBody:1.1"/>
    </coding>
    <text value="Documents following C-CDA constraints using a structured body."/>
  </payloadType>
  <payloadType>
    <coding>
      <system value="http://hl7.org/fhir/ValueSet/endpoint-payload-type"/>
      <code value="urn:hl7-org:sdwg:ccda-nonXMLBody:1.1"/>
    </coding>
    <text value="Documents following C-CDA constraints using a non structured body."/>
  </payloadType>
  <payloadMimeType value="text/xml"/>
  <payloadMimeType value="text/plain"/>
  <payloadMimeType value="application/pdf"/>
  <address value="dhol@directaddress.net"/>
</Endpoint>
```

24.3 Infrastructure Exchange: Non-FHIR

File: site/apis/24-Infrastructure-Exchange/3-Non-FHIR/index.html

* GetEncoderMessage

* HL7v2

* SetEncoderMessage

24.3.1 GetEncoderMessage

File: site/apis/24-Infrastructure-Exchange/3-Non-FHIR/700-GetEncoderMessage.html
Epic API ID | 700
FHIR version | Non-FHIR
Categories | Infrastructure Exchange
API type | Epic Public
API group | Encoder Group
USCDI | No
HTTP method | GET
URL template | api/epic/2016/Billing/Encoder/GETENCODERMESSAGE/EncoderMessage
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=700

Response elements

Name | Type | Optional | Array | Cardinality | Description
Error | Error | True
Error.Code | String | False
Error.Description | String | False
Message | String | REQUIRED if Error is null/omitted.

Errors

Code | Severity | Description | Example
MESSAGE-BUILD-FAILED | None | | The system failed to build the outgoing HL7v2 message.
SESSION-LOCK-FAILED | None | | Unable to obtain a lock on the coding session.

Sample request

```
GET https://hostname/instance/api/epic/2016/Billing/Encoder/GETENCODERMESSAGE/EncoderMessage HTTP/1.1
```

Sample response

```
<GETENCODERMESSAGEResponse xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
    xmlns="http://schemas.datacontract.org/2004/07/Epic.Billing.Generated.Models">
  <Error i:nil="true"/>
  <Message>MSH|^~\&amp;|EPIC|EPIC|||20160912122646||ADT^A08|21||2.3||||||||||EVN|A08|20160912122646|||19481^YUN^CHRIS^^
    PID|||115885^^^EPI^EPI||YUN^BEN||19960104|M|||123 ANYWHERE ST^^^^^USA^|||||||62913||||||||||||||PV1||I|^5988^61805^^
    ^^^^^^|KDE|||802266^YUN^CHRIS^^|||1|||||||802266^YUN^CHRIS^^|1||4|||||||||||||||01|||||In
    Progress|||20160223180703|20160223181300||0.00</Message>
</GETENCODERMESSAGEResponse>
```

24.3.2 HL7v2

File: site/apis/24-Infrastructure-Exchange/3-Non-FHIR/891-HL7v2.html
Epic API ID | 891
FHIR version | Non-FHIR
Categories | Infrastructure Exchange
API type | Interface
USCDI | No
HTTP method | POST
URL template | api/epic/2015/EDI/HTTP/HL7V2/{AIP}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=891

Description

This web service accepts an HL7v2 interface message payload. The webservice performs basic validation and can queue the message to a Bridges HL7 Interface in Epic.

For more information regarding HL7 over HTTP, refer to the HAPI Specification and the HL7v2 Interfaces tutorial.

The HTTP header should contain a Content-Type of x-application/hl7-v2+er7.

Epic replies with an HTTP 200 containing an HL7 Acknowledgement (ACK) or Negative Acknowledgement (NACK).

Request

Name | Type | Optional | Array | Description
ID | | False | | The interface ID for the interface to which the message should be routed.

Errors

Code | Severity | Description | Example
400 – Bad Request | None | | The embedded HL7 payload could not be validated. Additional information is logged to Trace.
401 | None | Unauthorized | The request failed basic authentication specified on the interface.
403 – Forbidden | None | | The interface requires HTTPS, but the request arrived on an http:// URL.
404 – Not Found | None | | The message was routed directly to an AIP, but the AIP does not exist or is inappropriate to route this message to. Additonal information is logged to Trace.
500 – Internal Server Error | None | | Additonal information is logged to Trace.

Sample request

```
MSH|^~\&|||YOUR APP|EPIC|MESSAGE DATE/TIME||MDM^T02|CONTROL ID|T|2.3
PID|||MRN>^^^^ID TYPE||LAST NAME^FIRST NAME^MIDDLE||DOB|LEGAL SEX||||||||||
PV1|1|||||||||||||||||EPIC VISIT ID (CSN)
TXA||DOCUMENT TYPE||ACTIVITY DATE/TIME|ACTIVITY PROVIDER ID^LAST^FIRST^MIDDLE^^^^^ID TYPE|||||||^^DOCUMENT ID|||||AU||AV
OBX|1|ED||1|^^PDF^^base 64 encoded document||||||||||
```

Sample response

```
MSH|^~\&|Sender|Receiver|||20200324132823-500||ACK|Control|T|2.3
MSA|AA|Control
```

24.3.3 SetEncoderMessage

File: site/apis/24-Infrastructure-Exchange/3-Non-FHIR/701-SetEncoderMessage.html
Epic API ID | 701

FHIR version | Non-FHIR
Categories | Infrastructure Exchange
API type | Epic Public
API group | Encoder Group
USCDI | No
HTTP method | POST
URL template | api/epic/2016/Billing/Encoder/SETENCODERMESSAGE/EncoderMessage
Supports OAuth 2.0 | True
Supported user types | Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=701

Request parameters

Name | Type | Optional | Array | Cardinality | Description
Error | Error | REQUIRED if Message is null/omitted.
Error.Code | String | False
Error.Description | String | False
Message | String | REQUIRED if Error is null/omitted.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Error | Error | True
Error.Code | String | False
Error.Description | String | False

Errors

Code | Severity | Description | Example
SESSION-LOCK-FAILED | None | | Unable to obtain a lock on the coding session.
MESSAGE-FILE-FAILED | None | | The system failed to file the incoming HL7 v2 message.

Sample request

```
POST https://hostname/instance/api/epic/2016/Billing/Encoder/SETENCODERMESSAGE/EncoderMessage HTTP/1.1
```

Sample response

```
<SETENCODERMESSAGERequest xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
    xmlns="http://schemas.datacontract.org/2004/07/Epic.Billing.Generated.Models">
  <Message>MSH|^~\&amp;amp;|EPIC|EPIC|||20160912122646||ADT^A08|21||2.3||||||||||EVN|A08|20160912122646|||19481^YUN^CHRIS^^
    PID|||115885^^^EPI^EPI||YUN^BEN||19960104|M|||123 ANYWHERE ST^^^^^USA^|||||||62913||||||||||||||PV1||I|^5988^61805^^
    ^^^^^^|KDE|||802266^YUN^CHRIS^^|||1|||||||802266^YUN^CHRIS^^|1||4|||||||||||||||01|||||In
    Progress|||20160223180703|20160223181300||0.00</Message>
</SETENCODERMESSAGERequest>
```